

FYP

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Let

$$X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}, \quad y = [y_1, \dots, y_n]^\top \in \{-1, +1\}^n.$$

The optimization variables are

$$D \in \mathbb{R}^{d \times m}, \quad C = [c_1, \dots, c_n] \in \mathbb{R}^{m \times n}, \quad w \in \mathbb{R}^m, \quad b \in \mathbb{R}, \quad u \in \mathbb{R}^n.$$

The ℓ_1 norm of C is defined as

$$\|C\|_1 = \sum_{i=1}^m \sum_{j=1}^n |c_{ij}|.$$

The pixel-wise box constraint on D is

$$0 \leq D_{ij} \leq 255, \quad \forall i, j,$$

which can be written using the indicator function

$$\delta_{[0,255]^{d \times m}}(D) = \begin{cases} 0, & D \in [0, 255]^{d \times m}, \\ +\infty, & \text{otherwise.} \end{cases}$$

0.1 Original Joint Objective

For each sample j , define the margin violation

$$r_j(C, w, b) = 1 - y_j(w^\top c_j + b).$$

The hinge loss is

$$\max(0, r_j(C, w, b)).$$

The original joint optimization problem is

$$\begin{aligned} \min_{C, D, w, b} \quad & \frac{1}{2} \|X - DC\|_F^2 + \frac{\alpha_1}{2} \|w\|_2^2 + \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, r_j(C, w, b)) \\ & + \lambda \|C\|_1 + \delta_{[0,255]^{d \times m}}(D). \end{aligned} \tag{1}$$

0.2 Auxiliary Variable and Quadratic Penalty

Introduce an auxiliary variable $u \in \mathbb{R}^n$.

For each j , define

$$q_j(C, w, b, u) = u_j - r_j(C, w, b) = u_j - (1 - y_j(w^\top c_j + b)). \tag{2}$$

Replace the hinge term by $\max(0, u_j)$ and enforce $u_j \approx r_j$ using a quadratic penalty with parameter $\alpha_0 > 0$:

$$\frac{\alpha_0}{2} \sum_{j=1}^n q_j(C, w, b, u)^2. \tag{3}$$

Impose the nonnegativity constraint

$$u \geq 0, \quad \delta_{\mathbb{R}_+^n}(u) = \begin{cases} 0, & u \in \mathbb{R}_+^n, \\ +\infty, & \text{otherwise.} \end{cases}$$

The penalized joint problem becomes

$$\begin{aligned} \min_{C,D,w,b,u} \quad & \frac{1}{2} \|X - DC\|_F^2 + \frac{\alpha_1}{2} \|w\|_2^2 + \frac{\alpha_0}{2} \sum_{j=1}^n q_j(C, w, b, u)^2 \\ & + \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j) + \lambda \|C\|_1 + \delta_{[0,255]^{d \times m}}(D) + \delta_{\mathbb{R}_+^n}(u). \end{aligned} \quad (4)$$

0.3 Composite Formulation

Define the smooth part

$$f(C, D, w, b, u) = \frac{1}{2} \|X - DC\|_F^2 + \frac{\alpha_1}{2} \|w\|_2^2 + \frac{\alpha_0}{2} \sum_{j=1}^n q_j(C, w, b, u)^2, \quad (5)$$

and the nonsmooth part

$$h(C, D, u) = \lambda \|C\|_1 + \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j) + \delta_{[0,255]^{d \times m}}(D) + \delta_{\mathbb{R}_+^n}(u). \quad (6)$$

The optimization problem is

$$\min_{C,D,w,b,u} f(C, D, w, b, u) + h(C, D, u).$$

0.4 Gradient Computation

Define

$$E = DC - X.$$

Then

$$\frac{1}{2} \|X - DC\|_F^2 = \frac{1}{2} \|DC - X\|_F^2 = \frac{1}{2} \|E\|_F^2.$$

Define

$$q = [q_1, \dots, q_n]^\top, \quad s = q \odot y.$$

Gradient with respect to C

The reconstruction term yields

$$\nabla_C \frac{1}{2} \|DC - X\|_F^2 = D^\top (DC - X) = D^\top E.$$

For the penalty term,

$$\nabla_{c_j} \frac{\alpha_0}{2} q_j^2 = \alpha_0 q_j \nabla_{c_j} q_j.$$

From

$$q_j = u_j - 1 + y_j (w^\top c_j + b),$$

we obtain

$$\nabla_{c_j} q_j = y_j w.$$

Thus

$$\nabla_{c_j} \frac{\alpha_0}{2} q_j^2 = \alpha_0 q_j y_j w.$$

Stacking columns gives

$$\nabla_C \left(\frac{\alpha_0}{2} \sum_{j=1}^n q_j^2 \right) = \alpha_0 w s^\top.$$

Therefore

$$\nabla_C f = D^\top E + \alpha_0 w s^\top.$$

Gradient with respect to D

$$\nabla_D f = (DC - X)C^\top = EC^\top.$$

Gradient with respect to w

$$\nabla_w \frac{\alpha_1}{2} \|w\|_2^2 = \alpha_1 w.$$

For the penalty term,

$$\nabla_w q_j = y_j c_j,$$

so

$$\nabla_w \frac{\alpha_0}{2} \sum_{j=1}^n q_j^2 = \alpha_0 \sum_{j=1}^n q_j y_j c_j = \alpha_0 C s.$$

Thus

$$\nabla_w f = \alpha_1 w + \alpha_0 C s.$$

Gradient with respect to b

$$\frac{\partial q_j}{\partial b} = y_j,$$

hence

$$\nabla_b f = \alpha_0 \sum_{j=1}^n q_j y_j = \alpha_0 \mathbf{1}^\top s.$$

Gradient with respect to u

$$\nabla_u f = \alpha_0 q.$$

0.5 Proximal Operators

C update

$$C^+ = \text{prox}_{\eta\lambda\|\cdot\|_1}(\tilde{C}) = \text{sign}(\tilde{C}) \odot \max(|\tilde{C}| - \eta\lambda, 0).$$

D update

$$D^+ = \text{clip}(\tilde{D}, 0, 255).$$

u update

Solve

$$\min_{u_j \geq 0} \frac{1}{2} (u_j - \tilde{u}_j)^2 + \eta \frac{\alpha_2}{2} u_j.$$

The unconstrained minimizer is

$$u_j^* = \tilde{u}_j - \eta \frac{\alpha_2}{2}.$$

Imposing $u_j \geq 0$ gives

$$u_j^+ = \max\left(0, \tilde{u}_j - \eta \frac{\alpha_2}{2}\right).$$

0.6 Joint Proximal Gradient Update

$$\tilde{z} = z^k - \eta \nabla f(z^k), \quad z^{k+1} = \text{prox}_{\eta h}(\tilde{z}).$$

0.7 Proximal Operators (Detailed Derivations)

In each proximal-gradient iteration, after the gradient step

$$\tilde{z} = z^k - \eta \nabla f(z^k),$$

the proximal step solves

$$z^{k+1} = \text{prox}_{\eta h}(\tilde{z}) \triangleq \arg \min_z \left\{ \frac{1}{2} \|z - \tilde{z}\|_2^2 + \eta h(z) \right\}.$$

Since

$$h(C, D, u) = \lambda \|C\|_1 + \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j) + \delta_{[0, 255]^{d \times m}}(D) + \delta_{\mathbb{R}_+^n}(u),$$

and each term depends on only one block among (C, D, u) , the proximal subproblem is separable:

$$\arg \min_{C, D, u} \left(\underbrace{\frac{1}{2} \|C - \tilde{C}\|_F^2 + \eta \lambda \|C\|_1}_{C\text{-block}} + \underbrace{\frac{1}{2} \|D - \tilde{D}\|_F^2 + \eta \delta_{[0, 255]^{d \times m}}(D)}_{D\text{-block}} + \underbrace{\frac{1}{2} \|u - \tilde{u}\|_2^2 + \eta \frac{\alpha_2}{2} \sum_j \max(0, u_j) + \eta \delta_{\mathbb{R}_+^n}(u)}_{u\text{-block}} \right).$$

Therefore,

$$C^+ = \text{prox}_{\eta \lambda \|\cdot\|_1}(\tilde{C}), \quad D^+ = \text{prox}_{\eta \delta_{[0, 255]}}(\tilde{D}), \quad u^+ = \text{prox}_{\eta(\frac{\alpha_2}{2} \sum \max(0, \cdot) + \delta_{\mathbb{R}_+^n})}(\tilde{u}),$$

and w, b are unchanged by the proximal step because h does not depend on (w, b) .

C update: ℓ_1 proximal map (soft-thresholding)

By definition,

$$C^+ = \text{prox}_{\eta \lambda \|\cdot\|_1}(\tilde{C}) = \arg \min_C \left\{ \frac{1}{2} \|C - \tilde{C}\|_F^2 + \eta \lambda \|C\|_1 \right\}.$$

Expand the Frobenius norm and the ℓ_1 norm elementwise:

$$\|C - \tilde{C}\|_F^2 = \sum_{i=1}^m \sum_{j=1}^n (c_{ij} - \tilde{c}_{ij})^2, \quad \|C\|_1 = \sum_{i=1}^m \sum_{j=1}^n |c_{ij}|.$$

Substituting yields

$$\frac{1}{2} \|C - \tilde{C}\|_F^2 + \eta \lambda \|C\|_1 = \sum_{i=1}^m \sum_{j=1}^n \left[\frac{1}{2} (c_{ij} - \tilde{c}_{ij})^2 + \eta \lambda |c_{ij}| \right].$$

Since the objective is a sum of independent functions of each scalar c_{ij} , the minimization decouples entrywise:

$$c_{ij}^+ = \arg \min_{c \in \mathbb{R}} \left\{ \frac{1}{2} (c - \tilde{c}_{ij})^2 + \eta \lambda |c| \right\}.$$

Let $a = \tilde{c}_{ij}$ and $\tau = \eta \lambda > 0$. Consider

$$\min_{c \in \mathbb{R}} \frac{1}{2} (c - a)^2 + \tau |c|.$$

Solve by cases.

Case 1: $c > 0$. Then $|c| = c$ and the objective is $\frac{1}{2}(c - a)^2 + \tau c$. Differentiating and setting to zero gives

$$(c - a) + \tau = 0 \quad \Rightarrow \quad c = a - \tau.$$

Feasibility ($c > 0$) requires $a > \tau$.

Case 2: $c < 0$. Then $|c| = -c$ and the objective is $\frac{1}{2}(c - a)^2 - \tau c$. Differentiating and setting to zero gives

$$(c - a) - \tau = 0 \quad \Rightarrow \quad c = a + \tau.$$

Feasibility ($c < 0$) requires $a < -\tau$.

Case 3: $c = 0$. When $a \in [-\tau, \tau]$, neither $a - \tau$ (positive) nor $a + \tau$ (negative) is feasible in its corresponding region. Equivalently, using the subgradient optimality condition at $c = 0$:

$$0 \in (c - a) + \tau \partial|c| \Big|_{c=0} \iff 0 \in -a + \tau[-1, 1] \iff a \in [-\tau, \tau].$$

Thus $c^+ = 0$ for $|a| \leq \tau$.

Combining cases yields the soft-thresholding operator:

$$c_{ij}^+ = \text{sign}(a) \max(|a| - \tau, 0) = \text{sign}(\tilde{c}_{ij}) \max(|\tilde{c}_{ij}| - \eta\lambda, 0).$$

Therefore, in matrix form,

$$C^+ = \text{sign}(\tilde{C}) \odot \max(|\tilde{C}| - \eta\lambda, 0),$$

where $\text{sign}(\cdot)$, $|\cdot|$, and $\max(\cdot, 0)$ act elementwise.

D update: projection onto the box $[0, 255]^{d \times m}$

By definition,

$$D^+ = \text{prox}_{\eta\delta_{[0, 255]^{d \times m}}}(\tilde{D}) = \arg \min_D \left\{ \frac{1}{2} \|D - \tilde{D}\|_F^2 + \eta\delta_{[0, 255]^{d \times m}}(D) \right\}.$$

Because the indicator is 0 on the feasible set and $+\infty$ otherwise, this is equivalent to Euclidean projection:

$$D^+ = \arg \min_{D \in [0, 255]^{d \times m}} \frac{1}{2} \|D - \tilde{D}\|_F^2.$$

Expand the Frobenius norm:

$$\|D - \tilde{D}\|_F^2 = \sum_{i=1}^d \sum_{j=1}^m (d_{ij} - \tilde{d}_{ij})^2,$$

and note the constraints are entrywise:

$$0 \leq d_{ij} \leq 255 \quad \forall i, j.$$

Thus the minimization decouples entrywise:

$$d_{ij}^+ = \arg \min_{0 \leq d \leq 255} \frac{1}{2} (d - \tilde{d}_{ij})^2.$$

The minimizer is the projection of $\tilde{d}_{ij}$ onto the interval $[0, 255]$:

$$d_{ij}^+ = \begin{cases} 0, & \tilde{d}_{ij} < 0, \\ \tilde{d}_{ij}, & 0 \leq \tilde{d}_{ij} \leq 255, \\ 255, & \tilde{d}_{ij} > 255, \end{cases}$$

which can be written compactly as

$$d_{ij}^+ = \min\{255, \max\{0, \tilde{d}_{ij}\}\}.$$

Therefore,

$$D^+ = \text{clip}(\tilde{D}, 0, 255).$$

u update: shifted ReLU from linear term + nonnegativity

The u -block proximal map is

$$u^+ = \arg \min_u \left\{ \frac{1}{2} \|u - \tilde{u}\|_2^2 + \eta \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j) + \eta \delta_{\mathbb{R}_+^n}(u) \right\}.$$

The indicator enforces $u \in \mathbb{R}_+^n$ (i.e., $u_j \geq 0$ for all j), so the problem is equivalent to

$$u^+ = \arg \min_{u \geq 0} \left\{ \frac{1}{2} \|u - \tilde{u}\|_2^2 + \eta \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j) \right\}.$$

Under the constraint $u \geq 0$, $\max(0, u_j) = u_j$ for all j , hence the objective becomes

$$u^+ = \arg \min_{u \geq 0} \left\{ \frac{1}{2} \|u - \tilde{u}\|_2^2 + \eta \frac{\alpha_2}{2} \sum_{j=1}^n u_j \right\}.$$

Expand the squared norm:

$$\|u - \tilde{u}\|_2^2 = \sum_{j=1}^n (u_j - \tilde{u}_j)^2,$$

so the objective separates across coordinates:

$$u^+ = \arg \min_{u \geq 0} \sum_{j=1}^n \left[\frac{1}{2} (u_j - \tilde{u}_j)^2 + \eta \frac{\alpha_2}{2} u_j \right],$$

which implies, for each j ,

$$u_j^+ = \arg \min_{u_j \geq 0} \left\{ \frac{1}{2} (u_j - \tilde{u}_j)^2 + \eta \frac{\alpha_2}{2} u_j \right\}.$$

Let $\gamma = \eta \frac{\alpha_2}{2} > 0$ and consider the scalar problem

$$\min_{u \geq 0} \frac{1}{2} (u - a)^2 + \gamma u, \quad a = \tilde{u}_j.$$

The unconstrained minimizer is obtained by differentiating and setting the derivative to zero:

$$\frac{d}{du} \left(\frac{1}{2} (u - a)^2 + \gamma u \right) = (u - a) + \gamma = 0 \quad \Rightarrow \quad u^* = a - \gamma.$$

Imposing $u \geq 0$ projects u^* onto $[0, \infty)$:

$$u^+ = \max(0, u^*) = \max(0, a - \gamma).$$

Substituting back yields

$$u_j^+ = \max \left(0, \tilde{u}_j - \eta \frac{\alpha_2}{2} \right), \quad \forall j.$$

0.8 Block Lipschitz Constants

We compute block-wise Lipschitz constants via the spectral norm of the Hessian:

$$L_x = \|\nabla_x^2 f\|_2.$$

With respect to C

Gradient:

$$\nabla_C f = D^\top (DC - X) + \alpha_0 w s^\top, \quad s_j = q_j y_j.$$

Reconstruction Hessian:

$$\nabla_C^2 \left(\frac{1}{2} \|DC - X\|_F^2 \right) = D^\top D.$$

Penalty Hessian (columnwise):

$$\nabla_{c_j}^2 \left(\frac{\alpha_0}{2} q_j^2 \right) = \alpha_0 w w^\top.$$

Thus

$$\nabla_C^2 f = D^\top D + \alpha_0 w w^\top.$$

Using subadditivity of the spectral norm:

$$L_C = \|\nabla_C^2 f\|_2 \leq \|D^\top D\|_2 + \alpha_0 \|w w^\top\|_2.$$

Since

$$\|w w^\top\|_2 = \|w\|_2^2,$$

we obtain

$$\boxed{L_C \leq \|D^\top D\|_2 + \alpha_0 \|w\|_2^2}.$$

With respect to D

Gradient:

$$\nabla_D f = (DC - X)C^\top.$$

Hessian:

$$\nabla_D^2 f = CC^\top.$$

Therefore

$$\boxed{L_D = \|CC^\top\|_2 = \|C\|_2^2}.$$

With respect to w

Gradient:

$$\nabla_w f = \alpha_1 w + \alpha_0 \sum_j q_j y_j c_j.$$

Hessian:

$$\nabla_w^2 f = \alpha_1 I + \alpha_0 \sum_j c_j c_j^\top.$$

Hence

$$L_w = \|\nabla_w^2 f\|_2 \leq \alpha_1 + \alpha_0 \left\| \sum_j c_j c_j^\top \right\|_2.$$

Since

$$\left\| \sum_j c_j c_j^\top \right\|_2 = \|C\|_2^2,$$

we obtain

$$\boxed{L_w \leq \alpha_1 + \alpha_0 \|C\|_2^2}.$$

With respect to b

Gradient:

$$\nabla_b f = \alpha_0 \sum_j q_j y_j.$$

Hessian:

$$\nabla_b^2 f = \alpha_0 n.$$

Thus

$$\boxed{L_b = \alpha_0 n.}$$

With respect to u

Gradient:

$$\nabla_u f = \alpha_0 q.$$

Hessian:

$$\nabla_u^2 f = \alpha_0 I.$$

Therefore

$$\boxed{L_u = \alpha_0.}$$

0.9 Step Size Rule

Define

$$L = \max\{L_C, L_D, L_w, L_b, L_u\}.$$

Choose

$$\boxed{\eta \leq \frac{1}{L}}.$$

0.10 Non-negativity constraint on u

For reference, the classification component of the soft-margin SVM can be written using the hinge loss

$$\ell_{\text{hinge}}(r_j) = \max(0, r_j),$$

where the margin residual is defined as

$$r_j = 1 - y_j(w^\top c_j + b).$$

The hinge loss is equivalent to introducing the slack variable

$$\xi_j = \max(0, r_j), \quad \xi_j \geq 0,$$

which measures the violation of the margin constraint

$$y_j(w^\top c_j + b) \geq 1.$$

When a sample lies outside the margin ($r_j < 0$), the hinge loss is automatically truncated to zero.

Auxiliary-variable formulation. To enable proximal-gradient optimization, we introduce an auxiliary variable u together with a quadratic penalty enforcing $u \approx r$:

$$\min_u \frac{\alpha_0}{2} \|u - r\|_2^2 + \frac{\alpha_2}{2} \sum_{j=1}^n \max(0, u_j).$$

Role of the non-negativity constraint. Without further constraints, the quadratic penalty drives u towards the residual r . In particular, when $r_j < 0$, the hinge term vanishes for $u_j < 0$, and the minimizer of the quadratic term is

$$u_j^* = r_j < 0.$$

Although the hinge loss remains zero in this case, the auxiliary variable becomes negative. This contradicts the interpretation of u_j as a slack variable, since slack variables must satisfy

$$\xi_j \geq 0.$$

To preserve the semantics of the hinge loss and ensure that the auxiliary variable behaves as a valid slack variable, we therefore impose the constraint

$$u \in \mathbb{R}_+^n.$$

With this constraint, the auxiliary variable satisfies

$$u_j \approx \max(0, r_j),$$

which is consistent with the slack-variable formulation of the soft-margin SVM.

0.11 Algorithm with Backtracking

Algorithm 1 Joint Proximal Gradient for Dictionary Learning and SVM

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1: Input:  $X, y, \alpha_0, \alpha_1, \alpha_2, \lambda$ , initial step size  $\eta_0$ , shrink factor  $\beta \in (0, 1)$ 
2: Initialize:  $C^0, D^0, w^0, b^0, u^0$ , define  $z^0 = (C^0, D^0, w^0, b^0, u^0)$ 
3: for  $k = 0, 1, 2, \dots$  until convergence do
4:   Compute residual  $E^k = D^k C^k - X$ 
5:   Compute margin mismatch  $q_j^k = u_j^k - (1 - y_j((w^k)^\top c_j^k + b^k))$ 
6:   Compute gradients
      $\nabla_C f = (D^k)^\top E^k + \alpha_0 w^k (s^k)^\top$ 
      $\nabla_D f = E^k (C^k)^\top$ 
      $\nabla_w f = \alpha_1 w^k + \alpha_0 \sum_j q_j^k y_j c_j^k$ 
      $\nabla_b f = \alpha_0 \sum_j q_j^k y_j$ 
      $\nabla_u f = \alpha_0 q^k$ 
     where  $s_j^k = q_j^k y_j$ .
7:   Set step size  $\eta = \eta_0$ 
8:   Backtracking line search
9:   repeat
10:    Gradient step
      $\tilde{C} = C^k - \eta \nabla_C f$ ,  $\tilde{D} = D^k - \eta \nabla_D f$ 
      $\tilde{w} = w^k - \eta \nabla_w f$ ,  $\tilde{b} = b^k - \eta \nabla_b f$ ,  $\tilde{u} = u^k - \eta \nabla_u f$ 
11:    Proximal updates
      $C^{trial} = \text{sign}(\tilde{C}) \odot \max(|\tilde{C}| - \eta \lambda, 0)$ 
      $D^{trial} = \text{clip}(\tilde{D}, 0, 255)$ 
      $u^{trial} = \max(0, \tilde{u} - \eta \frac{\alpha_2}{2})$ 
      $w^{trial} = \tilde{w}$ ,  $b^{trial} = \tilde{b}$ 
12:    Define  $z^{trial} = (C^{trial}, D^{trial}, w^{trial}, b^{trial}, u^{trial})$ 
13:    Sufficient decrease test
      $F(z^{trial}) \leq F(z^k) - \frac{1}{2\eta} \|z^{trial} - z^k\|_2^2$ 
14:    if test fails then
15:       $\eta \leftarrow \beta \eta$ 
16:    end if
17:  until test holds
18:  Accept update  $z^{k+1} = z^{trial}$ 
19: end for
20: Output:  $C, D, w, b, u$ 

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0.12 Newton-Type Acceleration for the Classifier Parameters

Newton Method. For a smooth function $f(x)$, the classical Newton update is defined as

$$x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k),$$

where $\nabla f(x_k)$ and $\nabla^2 f(x_k)$ denote the gradient and Hessian of the objective function. This update is obtained by minimizing the second-order Taylor approximation

$$f(x) \approx f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2} (x - x_k)^\top \nabla^2 f(x_k) (x - x_k).$$

Newton's method typically exhibits quadratic convergence when the Hessian is well conditioned and the iterates are sufficiently close to the optimum. However, computing and inverting the Hessian matrix can be computationally expensive for high-dimensional problems.

Proximal Newton Method.

$$F(x) = f(x) + h(x),$$

where f is smooth and h is possibly non-smooth. In this setting, proximal Newton methods extend Newton's method by incorporating the non-smooth term through a proximal operator. At iteration k , the update is obtained by solving

$$x_{k+1} = \arg \min_x \left(\nabla f(x_k)^\top (x - x_k) + \frac{1}{2} (x - x_k)^\top H_k (x - x_k) + h(x) \right),$$

where H_k is typically the Hessian $\nabla^2 f(x_k)$ or a positive definite approximation. This approach can significantly accelerate convergence compared with proximal gradient methods, although solving the subproblem may still be computationally demanding.

Quasi-Newton Methods. To avoid explicitly computing the Hessian matrix, quasi-Newton methods construct an approximation of the Hessian using gradient differences. A widely used update is the BFGS formula. Let

$$s_k = x_{k+1} - x_k, \quad y_k = \nabla f(x_{k+1}) - \nabla f(x_k).$$

parameter change and gradient change

The BFGS approximation of the Hessian matrix is updated as

$$H_{k+1} = H_k + \frac{y_k y_k^\top}{y_k^\top s_k} - \frac{H_k s_k s_k^\top H_k}{s_k^\top H_k s_k}.$$

The corresponding search direction is

$$x_{k+1} = x_k - H_k^{-1} \nabla f(x_k).$$

Quasi-Newton methods often achieve convergence speeds close to Newton's method while avoiding the explicit computation of second derivatives.

Limited-Memory BFGS (L-BFGS). For large-scale problems, storing the full Hessian approximation may still be impractical. The limited-memory BFGS (L-BFGS) method addresses this issue by storing only a small number of recent pairs (s_k, y_k) instead of the full matrix H_k . This greatly reduces memory requirements while preserving the key advantages of quasi-Newton methods.

0.12.1 Applicability to the Joint Dictionary–SVM Model

In the proposed joint model, the optimization variables are

$$C \in \mathbb{R}^{k \times n}, \quad D \in \mathbb{R}^{m \times k}, \quad w \in \mathbb{R}^k, \quad b \in \mathbb{R}, \quad u \in \mathbb{R}^n.$$

The matrices C and D contain a large number of parameters. Computing their Hessians would require matrices of size $(kn) \times (kn)$ and $(mk) \times (mk)$ respectively, which is computationally impractical. Consequently, these variables are updated using proximal gradient steps with backtracking line search as described in Algorithm 1.

In contrast, the classifier parameters (w, b) are low-dimensional. Therefore, applying a Newton-type update to this block is computationally feasible and can accelerate convergence of the classification component.

0.12.2 L-BFGS Update for the Classifier Parameters

Let

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix}.$$

The smooth part of the objective involving θ is

$$f(\theta) = \frac{\alpha_1}{2} \|w\|_2^2 + \frac{\alpha_0}{2} \sum_j q_j^2,$$

where

$$q_j = u_j - (1 - y_j(w^\top c_j + b)).$$

The gradient of f with respect to θ is

$$\nabla_{\theta} f = \begin{bmatrix} \nabla_w f \\ \nabla_b f \end{bmatrix},$$

with

$$\nabla_w f = \alpha_1 w + \alpha_0 \sum_j q_j y_j c_j, \quad \nabla_b f = \alpha_0 \sum_j q_j y_j.$$

Instead of computing the exact Hessian, the L-BFGS algorithm constructs a curvature approximation using the differences

$$s_k = \theta_{k+1} - \theta_k, \quad y_k = \nabla f(\theta_{k+1}) - \nabla f(\theta_k).$$

Using these quantities, the search direction is computed through the L-BFGS recursion, yielding the quasi-Newton update

$$\theta_{k+1} = \theta_k - H_k^{-1} \nabla f(\theta_k).$$

– $\mathcal{I}$ can meet the problem of splitting into 2 block A and B, A is for (C,D,u) and B is for (w,b) using L-BFGS, which becomes a block coordinate descent not updated together

0.13 Benchmark

Raw SVM – SVM on original features

Dictionary + SVM – two-stage pipeline

Joint Dictionary + SVM – proposed method